

Fungicide Efficacy—Biocontrols And Natural Products

Fungicide	Resistance risk (FRAC group number) ^{1,3}	Scab		Powdery mildew
		protectant	eradicant	
Actinovate	low, BM 02	2/3	0	2/3
Blight Ban**	low, BM 02	2/3	0	2/3
Blossom Protect	low, BM 02	2/3	0	2/3
Copper ²	low, M 01	2/3	0	2
Double Nickel 55	low, BM 02	2/3	0	2/3
Gargoil	low, BM 01	2	0	3
Lime sulfur	low, M 02	3	0	4
Procidic	low, BM 01	ND	0	ND
Regalia	low, P 05	2/3	0	2/3
Serenade	low, BM 02	2/3	0	2/3
Sulfur ²	low, M 02	3	0	5

Rating: 5 = excellent and consistent, 4 = good and reliable, 3 = moderate and variable, 2 = limited and/or erratic, 1 = minimal and often ineffective, 0 = ineffective, ND = no data.

**	Not registered, label withdrawn or inactive in California.
1	Group numbers are assigned by the Fungicide Resistance Action Committee(http://www.frac.info/) (FRAC) according to different modes of action (for more information, see www.frac.info). Fungicides with different group numbers are suitable to alternate in a resistance management program. In California, make no more than one application of a fungicide with a mode-of-action group number associated with high resistance risk before rotating to a fungicide with a different mode-of-action group number; for other fungicides, make no more than two consecutive applications before rotating to fungicide with a different mode-of-action group number.
2	Though copper may be effective for scab and blight control under low disease pressure, copper products may cause fruit scarring or russetting.
3	Biologicals: FRAC groups are provided as BM-No. or P-No. groups. In general, sulfur compounds are fungicidal and may affect applications of fungal biocontrols (e.g., Blossom Protect); whereas copper may affect applications of bacterial biocontrols (e.g., Actinovate, Bloomtime Biological, Blight Ban, Double Nickel 55, and Serenade). Rotations must consider these factors.

Acknowledgment: Adaskaveg et al., 2025. [Fungicides, Bactericides, Biocontrols, and Natural Products for Deciduous Tree Fruit and Nut, Citrus, Strawberry, and Vine Crops in California\(/PDF/PMG/fungicideefficacytiming.pdf\)](#).



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